

Day 8 Plan

Full Mock 1 + analysis · Physics revision (Days 1–7) · Current affairs re-test

Today's schedule (about 5 hours)

#	Block	Time	What you do
0	Recall	15 min	Yesterday's 10 questions (below) + redo yesterday's mistakes
1	FULL MOCK 1	90 min	Exam conditions — see the steps below
–	Break	15 min	Away from the screen
2	Analysis	60–90 min	The 6 steps below. Not optional.
3	Redo	30 min	Every wrong and skipped question, untimed
–	Break	10 min	Away from the screen
4	Physics revision	60 min	Read the one-page sheet (15 min) → 40-question rapid-fire quiz, target 12 min → re-read only the lessons behind your misses
5	Current affairs	20 min	20 MCQs from last month's part of your CA PDF — a re-test, not new material

Block 0 — Day 7 recall (answers on the last page)

#	Question	Your answer
1	Ohm's law?	
2	4 Ω and 4 Ω in parallel = ?	
3	Heater coils are made of?	
4	A 2 kW geyser for 1 h a day for 30 days uses how many units?	
5	Fuse wire is connected in which wire?	
6	Who discovered electromagnetic induction?	
7	Does a transformer work on DC?	
8	Who discovered X-rays?	
9	Maths: average 40 for 10 numbers, 64 read as 46 — correct average?	
10	Reasoning: in the morning your shadow falls to your right. Which way are you facing?	

Full Mock 1

Step	What to do						
Before	Testbook → an RRB JE CBT-I full previous paper you have not attempted (2026 shifts first, then Dec 2024). Phone away, water, rough sheets, a clock.						
During	90 minutes, one sitting, no pause. Practise the plan's order (GA → Science → Reasoning → Maths) and the two-round method. Mark every question you skip.						
Rule	No blind guesses. Guess only after eliminating at least one option.						
After	Fill the table below immediately, before you look at any solution.						
Mock 1	Attempted	Correct	Wrong	Left	Net	Time (min)	Target net
Maths							24.67
Reasoning							21.67
General Science							21.00
General Awareness							9.33
Total							76.67

Step	Analysis (60-90 min)
1	Net = correct – wrong ÷ 3, for each section.
2	Tag every wrong answer: concept / formula / calculation / misread / careless / guessed . Write the tags in your mistake log.
3	Solve every skipped question untimed. Solved in under 90 sec? → a selection or speed loss : you knew it. List these separately.
4	Time per section vs budget (GA 8 · Science 22 · Reasoning 22 · Maths 32 · review 6). Which one overran?
5	The 3 topics that cost the most marks → tomorrow's weak-topic drill.
6	Pick 5 questions to redo in 3 days. Send me the table + wrong and skipped questions — I'll diagnose them and send 5 similar ones.

Physics revision — the whole block on one page

Topic	Must remember
Units	SI base units: m, kg, s, A, K, mol, cd · light year = distance · 1 kWh = 3.6×10^6 J · 1 HP = 746 W · knot = speed · $^{\circ}\text{F} = 9/5 \text{ }^{\circ}\text{C} + 32$ (–40 is equal)
Motion	$v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · area under v-t = displacement · slope of v-t = acceleration · equal distances: $2xy/(x + y)$
Force	$F = ma$ · impulse = $F\Delta t$ = change in momentum · 1st law = inertia · 3rd law: rocket, gun recoil, swimming · mass = measure of inertia
Work, energy	$W = Fs \cos \theta$ · $KE = \frac{1}{2}mv^2$ (v doubled → KE × 4) · $PE = mgh$ · $P = W/t$ · energy is conserved
Gravitation	$F = Gm_1m_2/r^2$ · $g = 9.8 \text{ m/s}^2$, most at the poles, least at the equator, zero at the centre · Moon weight = 1/6 · escape velocity 11.2 km/s · geostationary: 24 h, ≈ 36,000 km
Fluids, heat	$P = F/A$ (pascal) · atmosphere ≈ 1.013×10^5 Pa = 76 cm Hg · Pascal → hydraulic lift · Archimedes → floating · Bernoulli → aeroplane lift · water densest at 4 °C
Sound	Needs a medium (not vacuum) · ≈ 343 m/s in air, faster in water and steel · audible 20 Hz–20 kHz · echo needs ≥ 17 m · pitch ↔ frequency, loudness ↔ amplitude

Topic	Must remember
Light	3×10^8 m/s · concave mirror: shaving, dentist, headlights · convex: rear-view · $P = 1/f$ (m) dioptre · myopia → concave lens, hypermetropia → convex · TIR: diamond, optical fibre, mirage · blue sky = scattering · red deviated least
Electricity	$V = IR$ · series: same current, R adds · parallel: same voltage (house wiring) · $P = VI = I^2R$ · 1 unit = 1 kWh · fuse (low melting point) in live wire · tungsten filament, nichrome heater
Magnetism	Oersted: current → magnetic field · Faraday: induction · Fleming left hand = motor, right hand = generator · transformer AC only, $V_s/V_p = N_s/N_p$

Physics rapid-fire — 40 questions, target 12 minutes (answers on the last pages)

#	Question	Answer
1	SI unit of luminous intensity?	
2	Light year is a unit of?	
3	1 kWh = ? J	
4	Temperature equal on °C and °F scales?	
5	Area under a velocity-time graph gives?	
6	A body starts from rest, $a = 2 \text{ m/s}^2$, $t = 5 \text{ s}$. Final velocity?	
7	Newton's first law is also called the law of?	
8	A rocket works on which law?	
9	Unit of impulse?	
10	Speed doubled → kinetic energy becomes?	
11	SI unit of power?	
12	Value of G?	
13	Where is g greatest on Earth?	
14	Weight on the Moon compared with Earth?	
15	Escape velocity from Earth?	
16	Period of a geostationary satellite?	
17	SI unit of pressure?	
18	Hydraulic lift works on?	
19	Aeroplane lift is explained by?	
20	Water is densest at?	
21	Why does ice float?	
22	Instrument to measure atmospheric pressure?	
23	Can sound travel in a vacuum?	
24	Audible range of humans?	
25	Minimum distance for an echo?	
26	Sound waves in air are longitudinal or transverse?	
27	Bats navigate using?	
28	Speed of light in vacuum?	
29	Mirror used in vehicle headlights?	
30	Mirror used as a rear-view mirror?	
31	Power of a lens with $f = 25 \text{ cm}$?	
32	Lens to correct myopia?	
33	Sparkle of a diamond is due to?	
34	Why is the sky blue?	
35	Colour deviated most by a prism?	

#	Question	Answer
36	Resistance of 3 Ω , 6 Ω in series?	
37	Household appliances are connected in?	
38	Filament of a bulb is made of?	
39	Rule for the direction of force in a motor?	
40	Device that steps voltage up or down?	

Answers

Day 7 recall

#	Answer	#	Answer
1	$V = IR$	6	Michael Faraday
2	$2\ \Omega$	7	No — AC only
3	Nichrome	8	Wilhelm Röntgen
4	60 units (kWh)	9	41.8
5	Live wire	10	South

Physics rapid-fire — 40 questions, target 12 minutes

#	Answer	#	Answer
1	Candela	21	It is less dense than water
2	Distance	22	Barometer
3	$3.6 \times 10^6\ \text{J}$	23	No
4	-40°	24	20 Hz – 20 kHz
5	Displacement	25	$\approx 17\ \text{m}$
6	10 m/s	26	Longitudinal
7	Inertia	27	Ultrasound
8	Newton's 3rd law	28	$3 \times 10^8\ \text{m/s}$
9	N·s (same as momentum)	29	Concave
10	4 times	30	Convex
11	Watt	31	+4 D (if convex)
12	$6.67 \times 10^{-11}\ \text{N m}^2/\text{kg}^2$	32	Concave
13	At the poles	33	Total internal reflection
14	1/6	34	Scattering of blue light
15	$\approx 11.2\ \text{km/s}$	35	Violet
16	24 hours	36	$9\ \Omega$
17	Pascal	37	Parallel
18	Pascal's law	38	Tungsten
19	Bernoulli's principle	39	Fleming's left-hand rule
20	$4\ ^\circ\text{C}$	40	Transformer

Send me tonight

Mock 1 table · your wrong and skipped questions · physics quiz score /40 · CA re-test /20.